

ELI5 Data Center Water use

Posted by u/Doc-Brown1911 • • 0 points • 20 comments

I don't understand why people are talking about data centers and water use so much. I'm not getting it.

Is it not a closed loop system? Yes they use a lot of water as a one time investment that only needs maintenance after that.

We don't hear people talking about how much water a nuclear reactor uses. Google says 11,000 to 60,000 gallons for every megawatt-hour (MWh) of electricity produced. plants withdraws roughly 11,000 to 60,000 gallons for every megawatt-hour (MWh) of electricity produced. That's a lot of water.

The only thing I can come up with is that the amount of data centers is going to dwarf the number of nuclear reactors.

Thoughts?

Comments

u/NonPartisanFinance • 1 points • 2025-12-23 14:12 UTC

Data centers need clean water which they usually pull from city and local water sources. So a big problem is the type of water used.

Nuclear, gas, coal, geothermal, power plants also use a ton of water! They also use a ton of water, but the water they pull from is less likely to be city water. Especially for the actual cooling water used vs a closed loop condensate system.

An important note here is that there is no 100% closed loop. All of these systems will have major losses due to evaporation. After all the whole goal of the water is to take the heat away from the servers and into the water. If you have enough heat you get steam and a lot of that you lose. This is why you see huge smoke stacks at power plants and small ones at data centers. That's basically all water vapor. And not harmful chemicals you assume they are. Those do get released, but are not visible as huge smoke stacks.

Another point is many locations are in severe droughts. They may not have enough water in their local water shed to cover the data center.

u/TheZerbio • 1 points • 2025-12-23 14:10 UTC

It depends in the Datacenter. Yes they need a Lot for initially filling the Loop and Afterwards there is some loss even in a closed Loop system. But AFAIK the bigger issue is evaporative cooling. In the hottest days right when the water supply is usually stressed anyway the closed Loop cooling can't keep up anymore since it gets less efficient with hotter temperatures outside.

That's when evaporative cooling gets activated. Which basically means they spray water on the cooling fins. As the water evaporates IT cools down. And this takes a lot of water at the worst possible time for the community.

u/windex_ninja • 1 points • 2025-12-23 14:07 UTC

I am going off memory so check my work.

The difference is that Data Centers use an evaporative cooling system for their servers which use millions of gallons a day and billions yearly which is more than most mid level towns of 50k people. Combine this with those same centers being built in already water scarce environments and you have a fight for resources driving up usage and cost.

u/GXWT • 1 points • 2025-12-23 14:07 UTC

> plants withdraws roughly 11,000 to 60,000 gallons for every megawatt-hour (MWh) of electricity produced

It's hard to approach this conversation in any meaningful matter if you're going to come up with some wild figures out of, what I can only assume is one of ChatGPT or your arse. I think you should do better research if you're going to try come up with demeaning comparisons. Several sources I find quote at least an order of magnitude less than this at either 1000s or even 100s.

u/explainlikeimfive-ModTeam • 1 points • 2025-12-23 14:06 UTC

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u/Cynical_Manatee • 0 points • 2025-12-23 14:06 UTC

It might eventually become a necessary evil like nuclear reactors, but right now it's hard to justify.

u/mattate • 1 points • 2025-12-23 14:06 UTC

So no, it's not a closed loop system.

u/ShinNefzen • 1 points • 2025-12-23 14:04 UTC

u/kaizen-rai • 2 points • 2025-12-23 14:04 UTC

u/greenmachine11235 • 1 points • 2025-12-23 14:04 UTC

u/penny-acre-01 • 6 points • 2025-12-23 14:03 UTC

u/gropula • 1 points • 2025-12-23 14:10 UTC

u/penny-acre-01 • 1 points • 2025-12-23 14:12 UTC

u/Doc-Brown1911 • 1 points • 2025-12-23 14:08 UTC

u/stpizz • 2 points • 2025-12-23 14:13 UTC

u/Doc-Brown1911 • 1 points • 2025-12-23 14:18 UTC

u/penny-acre-01 • 1 points • 2025-12-23 14:10 UTC

u/vikramdesh1 • 8 points • 2025-12-23 14:02 UTC

u/dudeplace • 1 points • 2025-12-23 14:31 UTC

u/NonPartisanFinance • 0 points • 2025-12-23 14:12 UTC

Yes, but also no. A closed loop system reuses a large portion of the water used for cooling, however you will always lose a significant amount of water to evaporation.

u/Unicorn_puke • 3 points • 2025-12-23 14:03 UTC

Some collect, but it's cheaper and more efficient to keep pumping in new water